

Digital Innovations

An AI Literacy Programme for the Age of Artificial Intelligence

COURSE OVERVIEW

Haileybury · 2026

Digital Innovations is Haileybury's artificial-intelligence literacy programme. It teaches pupils not just *how to use* AI tools, but how to **understand, question and apply them responsibly**. The whole course runs through an interactive web app — nothing to install, works offline, and free to operate — across **three pathways** tailored to different ages and goals, from a self-directed introduction to a full Academic Extension Programme with an independent build project.

Three pathways, one programme

Pathway	Best for	Lessons	What it delivers
Digital Innovations AEP	Sixth-Form & academic extension	59 · 6 units	The complete programme — theory, ethics, hands-on skills and an independent AI build, assessed by presentation & viva
Removes	Younger pupils (Removes year)	34 · 6 units	AI literacy and smart study skills, in short, friendly lessons
AI Fluency Track	Any pupil or staff member, self-paced	20 · 5 stages	A no-login, self-directed route to confident, critical everyday AI use

What pupils gain

- Understand how AI really works — from how chatbots predict text to what reasoning models and AI agents can (and cannot) do
- Write effective prompts and use AI as a genuine **thinking partner**, not a shortcut
- Critically evaluate and fact-check AI output, and spot deepfakes and misinformation
- Weigh the **ethics** — bias, privacy, environmental and human costs, copyright and AI companions
- Engage with policy and regulation, and form their own evidence-based positions
- **(AEP)** Design, build, ethically audit and pitch their own AI-powered project

Inside each pathway

Digital Innovations AEP — 59 lessons, 6 units

Foundations of AI (11) · Prompt Engineering & Research (8) · AI & Society (10) · Policy & Governance (5) · Capstone Project (16) · Presentations & Reflection (9)

Removes — 34 lessons, 6 units

Understanding AI (5) · AI for Study & Revision (7) · Practical AI Skills (7) · AI, Truth & Media (5) · AI, Society & Ethics (6) · AI, Wellbeing & Your Future (4)

AI Fluency Track — 20 lessons, 5 stages

Understanding AI (5) · Using AI Effectively (4) · Thinking Critically (3) · AI & Society (5) · Your AI Future (3)

The Fluency lessons are a curated selection drawn from the full AEP course.

A flavour of what we cover

Lessons tackle real, current questions — not abstractions. A sample of what pupils actually grapple with:

How AI really works

The Parrot vs. the Librarian — does AI understand, or just predict the next word?

Tokens & the Strawberry Problem — why AI miscounts the letters in a word, and what that reveals

Reasoning models — o3, GPT-5-thinking, Claude & DeepSeek R1: thinking before they answer

Agents, function calling & MCP — when AI stops chatting and starts *doing*

The big ethical questions

The Replika “apocalypse” — when a company changed millions of users' AI companions overnight

Regulating AI companions — after the death of a 14-year-old user, how would *you* legislate?

The hidden costs — nuclear-powered data centres, and workers paid \$1.32/hr to label trauma

NYT v OpenAI — who owns what an AI learned from?

Using AI with skill

The PTFC framework — Persona, Task, Format, Context: prompting that actually works

Trust, but verify — the four kinds of AI hallucination and how to catch them

Prompt injection — the No. 1 security flaw in AI applications

RAG — grounding AI in real documents instead of guesswork

AI, society & your future

Deepfakes, surveillance & facial recognition — synthetic media vs. civil liberties

AI & democracy — micro-targeting, influence operations and elections

Thinking With vs. Thinking For — what cognitive offloading really costs you

The capstone — design, build, ethically audit & pitch your own AI tool, defended at viva

How it works

- ✓ Runs in any web browser — **nothing to install**
- ✓ Works fully **offline** once loaded, and is **free** to operate
- ✓ Optional secure sign-in with the pupil's **school Google account**; progress syncs across devices

- ✓ Pupils earn XP, streaks and shareable **certificates**
- ✓ **Teacher dashboard** tracks completion & engagement — private notes stay private
- ✓ Glossary, live AI news, worked exemplars, quizzes & downloadable resources built in